# National University of Computer and Emerging Sciences
Islamabad Campus

**Technology Management (MG2007)**

# Sessional-I Exam

**Course Instructor(s):**
M. Hammad Majeed
**Section(s): (if applicable)**

**Total Time (Hrs):** 1
**Total Marks:** 15+45
**Total Questions:** 1+3

**Date:** Sep 23, 2025

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**Roll No** **Course Section** **Student Signature**

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**Attempt all the questions.**
**PART – I (15-minutes)**

**[CLO 1: Explain the strategic role of technology and innovation management in achieving organizational competitiveness**
**CLO 2: Analyze the frameworks and processes involved in planning, implementing, reviewing, and controlling innovation initiatives.**
**CLO 4: Apply appropriate tools and techniques for developing organizational capabilities in managing technology and fostering innovation.]**

**Q1:** This question consists of 20-MCQs. Mark your choice for each of the MCQ in the corresponding cell of table. Cutting/Overwriting will cancel the marks. **[15 marks]**

**1.** The fundamental strength of the Stage-Gate process lies in its ability to:
A) Increase bureaucratic control
**B) Enable structured decision-making across innovation phases**
C) Replace market testing
D) Minimize cross-functional collaboration
| **B** | The Stage-Gate model introduces "go/kill" gates to evaluate technical, market, and financial viability before resource commitment. |

**2.** A key failure in many traditional innovation projects occurs when:
A) Teams overcommunicate
B) Gate reviews lack cross-functional representation
C) Gates are skipped for speed
D) Both B and C
| **D** | Skipping or underpopulating gates eliminates critical checks from marketing, engineering, and finance — a common reason 40–50% of innovations fail. |

**3**. In the Waterfall SDLC, the greatest limitation for innovation projects is:
A) Predictable phase transition
B) Lack of user feedback during development
C) Iterative design testing
D) Continuous deployment
| **B** | Waterfall enforces sequential phases, limiting adaptation to evolving requirements. |

**4**. The Critical Path Method (CPM) is particularly useful in:
A) Estimating innovation revenue
B) Identifying non-sequential project risks
C) Determining the minimum completion time for interdependent tasks
D) Building agile backlog priorities
| **C** | CPM maps dependent activities to reveal the path with zero slack — controlling schedule risk. |

**5**.Microsoft's Responsible AI integration into Stage-Gate primarily redefines:
A) Technical feasibility reviews
B) Ethical compliance as a formal gate checkpoint
C) Budget approval at each stage
D) Product-market fit validation
| **B** | Microsoft introduced responsible AI "ethics gates" as non-negotiable review points in AI product lifecycles. |

**6**. Cooper's "Next-Generation Stage-Gate" differs from the original by:
A) Emphasizing more documentation
B) Integrating digital dashboards and iterative learning loops
C) Reducing feedback frequency
D) Eliminating gatekeepers
| **B** | Cooper's modernized model allows real-time metrics, flexibility, and agile iteration within traditional gating. |
**7**. A hybrid "Stage-Gate + Agile" approach improves innovation throughput by:
A) Adding bureaucracy to agile teams
B) Embedding iterative sprints within gate stages
C) Forbidding iteration within gates
D) Replacing all gates with sprints
| **B** | Agile cycles operate within gate-defined phases, combining adaptability with strategic oversight. |
**8**. In traditional Stage-Gate systems, "Stage 3: Development" typically focuses on:
A) Concept generation
B) Prototype design and scale-up
C) Full market launch
D) Idea screening
| **B** | Development stage translates the concept into a working prototype ready for testing. |
**9**. The Agile Manifesto values:
A) Tools over individuals
B) Processes over adaptability
C) Customer collaboration over contract negotiation
D) Documentation over execution
| **C** | Agile prioritizes customer interaction, adaptability, and working solutions over rigid contracts. |
**10**. A key differentiator between Scrum and Kanban is that Scrum:
A) Operates without iterations
B) Uses fixed-length sprints and defined roles
C) Focuses on continuous flow
D) Has no product owner role
| **B** | Scrum defines sprints, ceremonies, and roles; Kanban is continuous and flow-based. |
**11**. The Product Owner's central responsibility in Scrum is to:
A) Approve code merges
B) Manage the product backlog and prioritize value delivery
C) Supervise developer performance
D) Document sprint reports
| **B** | The Product Owner ensures backlog items align with business value and stakeholder needs. |
**12**. Continuous Integration (CI) enforces discipline by:
A) Allowing deployment without testing
B) Ensuring code passes automated tests before merging
C) Combining builds manually
D) Replacing QA teams
| **B** | CI integrates and tests code automatically, preventing faulty builds from progressing. |
**13**. Continuous Deployment (CD) enhances innovation speed by:
A) Automating the release process once tests pass
B) Reducing test coverage
C) Removing human oversight
D) Slowing down release cycles
| **A** | CD automates deployment after successful testing, allowing frequent, stable releases. |
**14**. Which of the following best describes DevOps philosophy?
A) Separation between dev and ops roles
B) Collaboration, automation, and shared ownership across delivery lifecycle
C) Rigid sequential delivery
D) Testing only after deployment
| **B** | DevOps eliminates silos between development and operations to enhance delivery speed and reliability. |
**15**. In DevOps, infrastructure as code (IaC) primarily aims to:
A) Manage IT infrastructure manually
B) Version-control infrastructure through code scripts
C) Limit automation
D) Simplify debugging only
| **B** | IaC uses scripts to automate setup, configuration, and scaling — maintaining consistency and auditability. |
**16**. The Spotify "Squad Model" illustrates:
A) Centralized decision-making
B) Autonomous, cross-functional teams (squads) organized into tribes and guilds
C) Hierarchical project management
D) Temporary committees
| **B** | Spotify decentralizes innovation through micro-units (squads) that function like mini-startups. |
**17**. Psychological safety in Agile teams is essential because:
A) It prevents iteration

B) It allows team members to voice ideas and failures without fear
C) It eliminates feedback loops
D) It enforces rigid compliance
| **B** | As per the HBR article, agile fails without psychological safety; openness drives iteration and innovation. |

**18**. A team consistently exceeding sprint velocity may indicate:
A) Underestimation of tasks
B) Overcommitment or unsustainable pace
C) Ideal productivity
D) Low backlog complexity
| **B** | Sustained overperformance signals unrealistic sprint planning and potential burnout. |

**19**. DevOps expands Agile principles primarily by:
A) Extending collaboration beyond development to include operations and deployment automation
B) Limiting iteration
C) Adding hierarchy
D) Reducing feedback frequency
| **A** | DevOps integrates Agile's iterative cycle with full-lifecycle automation and monitoring. |

**20**. Which statement about hybrid Agile–Stage-Gate systems is **most accurate**?
A) They are incompatible in high-tech projects
B) They combine structured governance with iterative execution
C) They eliminate project reviews
D) They rely entirely on waterfall principles
| **B** | Hybrid models provide governance checkpoints (gates) while enabling agile flexibility between them. |

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**PART – II (45 minutes)**

**[CLO 1: Explain the strategic role of technology and innovation management in achieving organizational competitiveness**
**CLO 4: Apply appropriate tools and techniques for developing organizational capabilities in managing technology and fostering innovation.]**

**Q1:** Netflix and Google represent two dominant but divergent innovation architectures.

- **Netflix** operates with a *"freedom and responsibility"* model emphasizing high trust, decentralized decision-making, and accountability. Teams manage their own budgets and timelines.
- **Google**, on the other hand, institutionalized *structural ambidexterity*: it allows employees to spend **20% of their work time on exploratory projects**, while core teams continue incremental innovation on flagship products (Search, Ads, Cloud).
  Both companies have faced challenges—Netflix's rapid scaling requires cultural coherence, while Google's "innovation at scale" struggles with bureaucratic friction.

**Your Task:**
Analyze how the two organizations' structures, cultures, and leadership philosophies support ambidextrous innovation. Use Schilling (Ch.10) and Steiber (Leadership for a Digital World) to evaluate how *team autonomy, cross-functional coordination,* and *reward mechanisms* affect innovation outcomes.
Support your analysis with one theoretical model (e.g., the Ambidextrous Organization, or Burns & Stalker's organic vs. mechanistic structures). **[15 marks]**

**Recommended Answer Summary:**

- Netflix = Cultural ambidexterity (trust + freedom) → organic structure enabling rapid decisions.
- Google = Structural ambidexterity (core vs. exploratory units).
- Theoretical link: Burns & Stalker's organic systems → adaptive, networked.
- Leadership: Steiber emphasizes *digital-age leadership* fostering *psychological safety* and *empowerment*.

- Integration outcome: Both balance *efficiency (exploit)* vs. *creativity (explore)* via culture–structure alignment.

**Mark Weightage:**

- Conceptual theory: 5 marks
- Application to Netflix & Google: 5 marks

Critical insight (ambidexterity, leadership): 5 marks

**[CLO 2: Analyze the frameworks and processes involved in planning, implementing, reviewing, and controlling innovation initiatives.**
**CLO 5: Assess the ethical, societal, and legal implications of deploying new technologies and formulate responsible innovation practices.]**

**Q2: Scenario:** Microsoft manages an expanding AI portfolio—Azure OpenAI, Copilot, Bing AI, and Responsible AI dashboards. Its innovation leadership faces a core dilemma:
How to **balance commercial acceleration** with **ethical governance** when deploying generative AI globally?

The AI unit's innovation committee uses **Stage-Gate ethics checkpoints**, **Agile roadmapping**, and **Balanced Scorecards** to monitor R&D outcomes. A tension exists between short-term market share pressure and long-term brand trust.

**Your Task:**
Design a **roadmap and portfolio structure** that manages this tension. Identify how the **BCG matrix** and **Agile roadmapping** guide resource allocation and ethical oversight. **[10 marks]**

**Recommended Answer Summary:**

- BCG Matrix Mapping:
  - *Stars*: Generative AI & Copilot features
  - *Cash Cows*: Azure AI services
  - *Question Marks*: Responsible AI models
  - *Dogs*: Deprecated cognitive APIs
- Agile Roadmap: Rolling-wave quarterly reviews; "Responsible AI" sprint integrated into all stages.
- Metrics: Ethical compliance %, model transparency audits, R&D ROI.
- Balance: Ethical AI as a **constraint**, not trade-off—trust sustains innovation value.
- Framework Link: Schilling (Ch.7) portfolio theory + Steiber's leadership for trust-based innovation ecosystems.

**Mark Weightage:**

- Theory integration (BCG, roadmapping): 6 marks
- Application to Microsoft: 6 marks
- Ethical dimension discussion: 3 marks

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**[CLO 3: Evaluate technology adoption models and their application in real-world organizational contexts.**
**CLO 4: Apply appropriate tools and techniques for developing organizational capabilities in managing technology and fostering innovation.**
**CLO5: Assess the ethical, societal, and legal implications of deploying new technologies and formulate responsible innovation practices.]**

**Q3: Scenario:** A leading automotive manufacturer is shifting toward electric and autonomous mobility. It maintains two innovation streams:

- **Exploitative** (incremental EV improvement)
- **Exploratory** (AI-driven autonomous systems).

Leadership struggles to develop KPIs that **encourage learning and experimentation** without penalizing short-term failure.

**Your Task:**
Propose an innovation performance framework using **Innovation Accounting** and **Balanced Scorecard principles** that aligns with an ambidextrous organization. Suggest both *leading* and *lagging* indicators. **[15 marks]**

**Recommended Answer Summary:**

- KPIs should emphasize **learning velocity**, **pivot rate**, and **R&D intensity**, not just ROI.
- Innovation Accounting (Ries): Measure learning progress via experiment validation, not revenue.
- Balanced Scorecard (Kaplan & Norton):
    - *Financial*: R&D ROI
    - *Customer*: Adoption rate
    - *Process*: Cycle time, defect rate
    - *Learning & Growth*: Experimentation per sprint, innovation satisfaction index.
- Outcome: Ambidexterity measured by dynamic balance—*efficiency + exploration.*

**Mark Weightage:**

- KPI design grounded in theory: 6 marks
- Practical integration to case: 6 marks
- Clarity & alignment with ambidexterity: 3 marks